The Structural Identity Switch of Planted Cliques: Collective Fitness Condensation in Bianconi–Barabási Networks

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Abstract

A k-clique planted in a Bianconi–Barabasi fitness network undergoes two sharp phase transitions as its collective fitness eta increases. Below a critical fitness eta_c(k) = 0.4879 + 0.5477/k, the clique is captured as a community by Louvain modularity detection (pcap_louvain approaches 1). Above eta_c, community capture collapses to zero. Above a second threshold eta_cross(k) = 0.6296 + 1.1554/k, the clique members rank as the top-k nodes by betweenness centrality (pcap_bc exceeds pcap_louvain). Between these two thresholds lies an identity crisis zone of width Delta_eta(k) = 0.1417 + 0.6077/k where neither metric dominates. Both transitions follow a + b/k scaling with finite asymptotes as k approaches infinity and are independent of network size n but shift upward with the attachment parameter m. The crossover at eta_cross is a structural identity switch: below it the clique is a community object, embedded passively in the network; above it the clique is a hub object, routing traffic through its members. This is collective fitness condensation, a k-body generalization of the single-node Bose–Einstein condensation identified in the original Bianconi–Barabasi model.

1 Introduction

The Bianconi–Barabási (BB) fitness model [Bianconi and Barabási, 2001b] assigns each node an intrinsic fitness parameter that modulates preferential attachment. A node with high fitness attracts edges faster than its degree alone would predict. When fitness is drawn from certain distributions, a single node can capture a macroscopic fraction of all edges, a phenomenon identified as Bose–Einstein condensation in complex networks [Bianconi and Barabási, 2001a]. This condensation is a single-node effect: one winner takes all. Moreover, the standard Barabási–Albert model without fitness does not produce emergent community structure [Vazquez, 2025]: BA($m = 2$) networks have no finite Ramsey community number r_κ under either stochastic block model or Infomap detection, confirming that pure degree-driven preferential attachment is an insufficient substrate for studying how growth dynamics govern community identity. The fitness extension of Bianconi and Barabási [2001b] is therefore the natural starting point.

Real networks contain structural motifs, not just isolated nodes. A team of collaborators in a citation network, a protein complex in a metabolic network, or a densely connected routing core in an infrastructure network is not a single node but a k-body object. Such motifs can share a collective advantage: they may all be early entrants, all be high-quality, or all benefit from mutual reinforcement. The natural question is what happens when a group of nodes, rather than a single node, shares a collective fitness advantage in a growing network.

This question has not been asked. The BB literature treats fitness as a node-level property [Bianconi and Barabási, 2001b,a, Ferretti and Bianconi, 2008]. The community detection literature asks whether planted structures can be recovered from observed networks [Fortunato, 2010, Decelle et al., 2011], but does not model how fitness-driven growth determines which structural role a planted motif occupies. The k-clique percolation literature [Palla et al., 2005, Derényi et al., 2005] establishes that a giant k-clique community percolates in an Erdős–Rényi graph when the edge probability crosses a critical threshold $p_c(k) = [(k-1)N]^{-1/(k-1)}$; our $\eta_c(k)$ is the fitness-space analog of this percolation threshold, identifying the critical collective fitness below which the planted clique is detected as a community in a BB network. The key distinction is that $p_c(k)$ is a static percolation condition on a fixed random graph, whereas $\eta_c(k)$ is a dynamical property intrinsic to fitness-driven growth.

We plant a k-clique in a BB network before growth begins and assign all clique members a shared fitness parameter η. As η increases, we track two capture metrics: the probability that Louvain community detection recovers the clique as a community ($p_{\text{cap}}^{\text{Louv}}$), and the probability that the clique members rank as the top-k nodes by betweenness centrality ($p_{\text{cap}}^{\text{BC}}$). These two metrics probe fundamentally different structural roles. A community is a group of nodes more densely connected to each other than to the rest of the network. A hub set is a group of nodes through which network traffic is routed. These roles are not the same, and the transition between them is not gradual.

Our contributions are:

1. **Phase boundary** $\eta_c(k)$: A sharp threshold below which the clique is a community object ($p_{\text{cap}}^{\text{Louv}} \approx 1$) and above which community capture collapses. The threshold follows $\eta_c(k) = 0.4879 \pm 0.0151 + (0.5477 \pm 0.0521)/k$.

2. **Crossover** $\eta_{\textbf{cross}}(k)$: A second, higher threshold above which betweenness centrality dominates ($p_{\text{cap}}^{\text{BC}} > p_{\text{cap}}^{\text{Louv}}$). Fit: $\eta_{\text{cross}}(k) = 0.6296 \pm 0.0149 + (1.1554 \pm 0.0513)/k$.

3. **Identity crisis zone**: A finite-width interval $\Delta\eta(k) = 0.1417 \pm 0.0293 + (0.6077 \pm 0.1008)/k$ between the two transitions where neither metric dominates. The zone narrows as $1/k$ but never closes.

4. **Structural identity switch**: At η_{cross}, the $p_{\text{cap}}^{\text{Louv}}$ and $p_{\text{cap}}^{\text{BC}}$ curves cross cleanly. Below the crossover the clique is a community; above it the clique is a hub. This is the central phenomenon.

5. **Universal** $a + b/k$ **scaling**: Both transitions and the dead-zone width follow the same functional form, with finite asymptotes as $k \to \infty$.

6. n**-independence**: Neither threshold shifts with network size ($n = 500, 1000, 2000$). These are intrinsic properties of the BB fitness dynamics.

7. m**-dependence**: Both thresholds shift upward with the BA attachment parameter m. A better-connected background raises the fitness bar for the clique.

The thesis is direct: structural motifs with sufficient collective fitness actively reshape the network around them. This is collective fitness condensation, a k-body generalization of the single-node BEC identified by Bianconi and Barabási [2001a].

Section 2 defines the model and measurements. Section 3 presents the seven findings. Section 4 connects to prior theory. Section 5 summarizes and states open questions.

2 Model and Methods

2.1 Bianconi–Barabási Fitness Network

The Bianconi–Barabási (BB) model [Bianconi and Barabási, 2001b] extends the Barabási–Albert preferential attachment model [Barabási and Albert, 1999] by assigning each node i an intrinsic fitness parameter η_i drawn from a distribution $\rho(\eta)$. At each time step, a new node arrives with m edges. Each edge attaches to an existing node j with probability proportional to $\eta_j \cdot k_j$, where k_j is the current degree of node j. In the standard BB model, $\rho(\eta)$ is the exponential distribution with mean 1:

$$\rho(\eta) = e^{-\eta}, \quad \eta \geq 0. \tag{1}$$

We use $m = 2$ as the default attachment parameter (each new node brings two edges) and grow the network to $n = 1000$ nodes unless stated otherwise.

2.2 Planted Clique with Collective Fitness

Before growth begins, we initialize k nodes as a complete subgraph (a k-clique). All $\binom{k}{2}$ edges among these k nodes are present at $t = 0$. Each clique member is assigned the same fitness parameter η, which we treat as the control parameter of the model. The clique members therefore participate in the BB preferential attachment process from the first time step, with attachment probability proportional to $\eta \cdot k_j$ for each clique member j.

The remaining $n - k$ nodes are added sequentially, each drawing its fitness from the exponential distribution (Eq. 1) independently. These background nodes attach via standard BB preferential attachment.

The planted clique has two advantages: temporal (its members are present from $t = 0$, accumulating degree throughout growth) and fitness-based (all members share η, which may exceed the mean background fitness of 1). The parameter η controls the relative strength of the fitness advantage. At $\eta = 0$ the clique members have zero attractiveness and gain no new edges during growth; at $\eta = 1$ they have the same expected fitness as background nodes; at $\eta \gg 1$ they dominate the attachment process.

2.3 Capture Metrics

We measure two capture probabilities as functions of η for each clique size k.

Community capture ($p_{\text{cap}}^{\text{Louv}}$). We apply the Louvain community detection algorithm [Newman and Girvan, 2004, Blondel et al., 2008] to the final network using the `python-louvain` package (function `community.best_partition`) with resolution parameter $\gamma = 1$ (the default). A replicate is scored as captured if all k clique members are assigned to the same detected community. The community capture probability is the fraction of replicates in which this condition holds:

$$p_{\text{cap}}^{\text{Louv}}(\eta, k) = \frac{1}{R} \sum_{r=1}^{R} \mathbf{1}[\text{all } k \text{ members in one Louvain module in replicate } r]\,. \tag{2}$$

Betweenness centrality capture ($p_{\text{cap}}^{\text{BC}}$). We compute betweenness centrality [Freeman, 1977, Goh et al., 2001] for all n nodes in the final network using NetworkX. A replicate is scored as captured if the k clique members are exactly the top-k nodes by betweenness centrality:

$$p_{\text{cap}}^{\text{BC}}(\eta, k) = \frac{1}{R} \sum_{r=1}^{R} \mathbf{1}[\text{top-}k \text{ BC nodes} = \mathcal{K} \text{ in replicate } r] \tag{3}$$

where $\mathcal{K}$ is the clique member set.

These two metrics probe different structural roles. Community capture asks whether the clique is a cohesive module, more densely connected internally than externally. Betweenness capture asks whether the clique members are routing hubs, positioned on the shortest paths between other nodes. The use of two independent metrics is motivated by prior work showing that single-metric assessments of planted clique influence are sensitive to measurement convention: the same clique can appear dominant or irrelevant depending solely on how coverage is measured [Edwards, 2026].

2.4 Parameter Space and Simulation Details

Primary sweep. The control parameter η is swept from 0.0 to 3.0 in increments of 0.05 (61 values). Clique sizes $k \in \{3, 4, 5, 6, 7, 8\}$ are tested. For each (η, k) pair, 500 independent replicates are generated. Network size is $n = 1000$ and attachment parameter is $m = 2$. Total simulation count for the primary sweep: $61 \times 6 \times 500 = 183{,}000$ networks.

n-scaling. Network sizes $n \in \{500, 1000, 2000\}$ are tested at $k \in \{3, 4, 5, 6\}$, $m = 2$, with 300 replicates per (η, k, n) combination.

m-scaling. Attachment parameters $m \in \{2, 3, 4\}$ are tested at $n = 1000$, $k \in \{3, 4, 5, 6\}$, with 300 replicates per (η, k, m) combination.

Threshold extraction. The phase boundary $\eta_c(k)$ is defined as the value of η at which $p_{\text{cap}}^{\text{Louv}}$ crosses 0.5 from above (interpolated linearly between adjacent grid points). The crossover $\eta_{\text{cross}}(k)$ is defined as the value of η at which $p_{\text{cap}}^{\text{BC}} = p_{\text{cap}}^{\text{Louv}}$ (interpolated linearly). Both are fit to the functional form $a + b/k$ using weighted least squares, with weights inversely proportional to the squared uncertainty at each k.

Implementation. All simulations are implemented in Python using NetworkX for graph construction, betweenness centrality computation, and the `python-louvain` package for Louvain community detection. Parallelization uses the `multiprocessing` module with process-level replication. Random seeds are drawn independently for each replicate.

3 Results

3.1 Phase Boundary $\eta_c(k)$: Collapse of Community Capture

At low η, clique members have weak fitness and attract few additional edges during growth. The $\binom{k}{2}$ internal clique edges remain the densest local structure, and Louvain reliably identifies the clique as a community module. Community capture probability $p_{\text{cap}}^{\text{Louv}}$ is near 1.0 for all k when $\eta < 0.3$.

As η increases past a critical value η_c, community capture drops sharply. The transition is not gradual: $p_{\text{cap}}^{\text{Louv}}$ falls from above 0.9 to below 0.1 over an η interval of approximately 0.2 for all k tested. The mechanism is straightforward. As clique members gain fitness, they attract more external edges. Each external edge weakens the modularity of the clique by increasing the ratio of external to internal connections [Fortunato and Barthélemy, 2007]: as clique members accumulate external connections, the modularity gain of maintaining the clique as a distinct module falls below the null-model expectation, and Louvain assigns members to their external neighbor communities. Above η_c, the clique members are so well-connected to the rest of the network that they are no longer a modularity-distinct group.

The critical fitness η_c depends on clique size. Larger cliques have more internal edges ($\binom{k}{2}$ scales quadratically) and therefore tolerate more external connections before modularity dissolves. The measured values of $\eta_c(k)$ for $k = 3, 4, 5, 6, 7, 8$ are well described by:

$$\eta_c(k) = 0.4879 \pm 0.0151 + \frac{0.5477 \pm 0.0521}{k} \tag{4}$$

with $R^2 > 0.99$ (Figure 2). The asymptote $\eta_c(\infty) = 0.4879$ represents the intrinsic fitness threshold of the BB model at $m = 2$: the fitness above which any planted structure, regardless of size, loses community identity.

3.2 Crossover $\eta_{\text{cross}}(k)$: Emergence of Hub Identity

While community capture collapses above η_c, betweenness centrality capture $p^{\text{BC}}_{\text{cap}}$ rises monotonically with η. At low η, clique members have no special position in the shortest-path structure of the network and $p^{\text{BC}}_{\text{cap}} \approx 0$. As η increases, clique members attract more edges, gain higher degree, and progressively insert themselves onto shortest paths between other nodes.

At a second threshold η_{cross}, the rising $p^{\text{BC}}_{\text{cap}}$ curve crosses the falling $p^{\text{Louv}}_{\text{cap}}$ curve. The crossing is clean: the two curves intersect at a single point for each k, with no ambiguity or gradual handoff (Figure 1). Below η_{cross}, the dominant capture metric is community detection; above it, the dominant metric is betweenness centrality. This crossover is the structural identity switch.

The crossover fitness is:

$$\eta_{\text{cross}}(k) = 0.6296 \pm 0.0149 + \frac{1.1554 \pm 0.0513}{k} \tag{5}$$

with $R^2 > 0.99$ (Figure 2). The asymptote $\eta_{\text{cross}}(\infty) = 0.6296$ is the fitness above which a sufficiently large planted clique inevitably becomes a hub object in the BB network.

The structural identity switch is the central result of this paper. It demonstrates that the same structural motif (a k-clique) occupies qualitatively different roles in the network depending on a single continuous parameter (collective fitness). The clique does not merely become "more important" with higher fitness. It changes what it is.

3.3 Identity Crisis Zone $\Delta\eta(k)$

Between η_c and η_{cross} lies a finite interval where the clique is neither a good community object (community capture has collapsed) nor a good hub object (betweenness capture has not yet risen to dominance). We call this the identity crisis zone.

The width of the zone is:

$$\Delta\eta(k) = \eta_{\text{cross}}(k) - \eta_c(k) = 0.1417 \pm 0.0293 + \frac{0.6077 \pm 0.1008}{k} \tag{6}$$

The zone is widest at small k: for $k = 3$, $\Delta\eta \approx 0.34$; for $k = 8$, $\Delta\eta \approx 0.22$. The $1/k$ correction narrows the zone as clique size grows, but the asymptote $\Delta\eta(\infty) = 0.1417$ is strictly positive. The identity crisis zone never closes. Even in the $k \to \infty$ limit, there exists a finite fitness interval where a planted clique is structurally ambiguous: it has lost its community identity but has not yet acquired hub identity.

Within this zone, both $p_{\text{cap}}^{\text{Louv}}$ and $p_{\text{cap}}^{\text{BC}}$ are below 0.5. The clique is present in the network but occupies no well-defined structural role. It is visible to neither standard community detection nor standard centrality ranking. This has practical implications: motifs in the identity crisis zone are structurally invisible to the two most common characterization tools.

3.4 Universal $a + b/k$ Scaling

All three quantities (η_c, η_{cross}, $\Delta\eta$) follow the functional form $a+b/k$ with high precision ($R^2 > 0.99$). The form has a physical interpretation. The constant a is the asymptotic threshold in the limit of infinite clique size, determined by the interplay between the BB fitness distribution and the attachment parameter m. The b/k correction represents the finite-size penalty of the planted clique: smaller cliques need higher per-member fitness to achieve the same collective effect because they have fewer internal edges and a smaller footprint in the network.

The $1/k$ dependence is not imposed by the fitting procedure. We tested $a + b/k^{\alpha}$ with α as a free parameter; the best-fit α is indistinguishable from 1.0 for all three quantities ($\alpha = 0.98 \pm 0.04$ for η_c, $\alpha = 1.01 \pm 0.03$ for η_{cross}, $\alpha = 1.03 \pm 0.06$ for $\Delta\eta$). The $a + b/k$ form is exact within measurement precision.

3.5 n-Independence of Thresholds

To test whether η_c and η_{cross} are finite-size artifacts, we repeated the primary sweep at $n = 500$, $n = 1000$, and $n = 2000$ for $k \in \{3, 4, 5, 6\}$ with $m = 2$ and 300 replicates per condition.

Neither threshold shifts with n. The measured values of $\eta_c(k)$ at $n = 500$, 1000, and 2000 agree within error bars for all four values of k tested (Figure 3, left panel). The same holds for $\eta_{\text{cross}}(k)$. The $a + b/k$ fit parameters are statistically identical across network sizes.

This n-independence is expected on physical grounds. The BB attachment probability is proportional to $\eta_j \cdot k_j / \sum_i \eta_i \cdot k_i$. As n grows, both the numerator (clique member attractiveness) and the denominator (total network attractiveness) scale with n, preserving the relative fitness advantage. The thresholds η_c and η_{cross} are intrinsic properties of the competition between the planted clique and the background fitness distribution, not properties of the network size.

3.6 m-Dependence of Thresholds

The attachment parameter m controls the baseline connectivity of the network. At $m = 2$, each new node brings two edges; at $m = 4$, four. Higher m produces a denser, better-connected background in which each node has more edges on average.

Both η_c and η_{cross} shift upward with m (Figure 3, right panel). At $m = 2$, $\eta_c(k = 5) \approx 0.60$; at $m = 4$, $\eta_c(k = 5) \approx 0.80$. The shift is systematic across all tested k values.

The mechanism is intuitive. A denser background means background nodes accumulate higher degrees during growth. The clique must have proportionally higher fitness to attract edges at the same rate relative to the background. The fitness bar for both transitions rises with m because the competition is stiffer.

The $a + b/k$ functional form is preserved at all m values; only the coefficients a and b change. This is consistent with the interpretation that a encodes the competition between planted fitness and the background fitness distribution, mediated by m, while b/k encodes the finite-size correction of the clique.

4 Discussion

4.1 Collective Fitness Condensation

Bianconi and Barabási [2001a] showed that in the BB model with certain fitness distributions, a single node can capture a macroscopic fraction of edges, analogous to Bose–Einstein condensation in quantum gases. That condensation is a single-node phenomenon: one node wins the competition.

Our results demonstrate a collective version of this effect. A k-clique with shared fitness η undergoes a qualitative structural transition as η increases, not by capturing all edges (the clique members do not monopolize attachment) but by reorganizing its structural role in the network. Below η_c the clique is passively embedded as a community module. Above η_{cross} the clique has become the network's routing backbone, with its members dominating betweenness centrality. The clique does not merely accumulate more edges; it actively reshapes the network topology around itself. The dynamics of this condensation process — specifically the timescale over which a motif achieves macroscopic structural dominance — have been characterized by Ferretti and Bianconi [2008], who show that this timescale diverges as fitness approaches the critical threshold from below. This divergence is consistent with the gradual, $\sim$0.2-wide transitions in η observed here: near η_c and η_{cross}, the system is in a critical slowing regime where dominance takes longer to establish, producing the observed finite-width crossovers rather than discontinuous jumps.

This is not a contradiction of single-node BEC but a generalization. Single-node condensation is the $k = 1$ limit of collective fitness condensation. What we add is the $k > 1$ structure: the phase boundary, the crossover, the identity crisis zone, and the $a + b/k$ scaling that connects the two regimes.

The analogy to BEC is structural, not thermodynamic. We do not claim a mapping to a partition function or a free energy. The parallel is that a continuous control parameter (fitness) drives a sharp transition in the macroscopic role of a microscopic object (a node or a clique), and that the transition is intrinsic to the growth dynamics rather than imposed by external manipulation.

4.2 Relation to the Stochastic Block Model Detectability Transition

Decelle et al. [2011] established a phase transition in the stochastic block model (SBM): below a critical signal-to-noise ratio, no algorithm can recover the planted community structure. This detectability transition is a fundamental limit on inference.

Our phase boundary η_c superficially resembles the SBM detectability threshold, but the mechanism and model class are entirely different. The SBM is a static random graph with planted structure; the question is whether an observer can detect the structure from the topology. The BB model is a growing network with fitness-driven dynamics; the question is what structural role the planted motif occupies in the grown network. The SBM threshold is an information-theoretic limit on inference. Our threshold η_c is a dynamical transition in the network's structure itself, not in our ability to detect it.

Put concretely: above η_c in our model, the clique is not hidden. It is structurally present (its members have high degree and high betweenness centrality). It has simply ceased to be a community. The Louvain algorithm does not fail to detect it; there is no community to detect. The clique has changed what it is, not how visible it is.

The conceptual parallel is real: both results show that a planted structure's detectability depends on the parameters of the generative model. But the mechanisms are orthogonal, and the models share no formal connection.

4.3 Physical Interpretation of the $a + b/k$ Form

The scaling $\eta_c(k) = a + b/k$ admits a simple physical reading. The constant $a = 0.4879$ is the fitness threshold at which a single clique edge begins to lose modularity advantage over the background. This threshold depends on the background fitness distribution (exponential with mean 1) and the attachment parameter m, but not on the clique size.

The b/k correction reflects the finite internal edge density of the clique. A k-clique has $\binom{k}{2}$ internal edges and k members, giving an internal degree of $k-1$ per member. As k grows, the internal degree grows linearly, providing a larger modularity buffer against external edge accumulation. The $1/k$ correction quantifies how much additional fitness the clique can tolerate before this buffer is overwhelmed. At $k \to \infty$, the buffer is infinite and only the intrinsic threshold a remains.

The same logic applies to η_{cross}, with a different constant $a = 0.6296$ because betweenness centrality capture requires a stronger condition than modularity loss. To dominate betweenness, the clique must not only shed its community structure but actively centralize the network's shortest paths through its members. This requires higher fitness and therefore a higher threshold.

4.4 The Identity Crisis Zone as an Empirical Prediction

The identity crisis zone is not merely an artifact of using two metrics. It is a predicted structural regime in any fitness-driven network containing planted or emergent motifs. In this regime, a motif exists in the network but is invisible to both community detection and centrality analysis. It has lost its modular cohesion but has not yet acquired routing dominance.

This prediction is empirically testable. In networks where node fitness can be estimated or controlled (citation networks with journal impact factors, social networks with influence scores, infrastructure networks with capacity parameters), one can identify structural motifs and ask whether their members cluster in a single community or rank as top centrality nodes. The identity crisis zone predicts a fitness range where motifs do neither, and where their structural role is ambiguous under standard analysis tools.

The zone width $\Delta\eta = 0.1417 + 0.6077/k$ provides a quantitative prediction. For a triad ($k = 3$), the zone spans $\Delta\eta \approx 0.34$ in fitness units. For an octad ($k = 8$), it narrows to $\Delta\eta \approx 0.22$. The prediction is falsifiable: if real networks show a clean, zero-width transition from community to hub with no intermediate regime, the identity crisis zone is an artifact of the BB model and does not generalize.

4.5 Implications for Network Analysis

The structural identity switch has a practical implication for network analysis methodology. Community detection and centrality analysis are routinely applied to the same network, often by different groups, producing complementary characterizations of the same nodes. Our results show that these two analyses can give contradictory answers about the same structural motif, and that the contradiction is not an error but a correct reflection of where the motif sits in the fitness-driven phase diagram.

A clique detected as a community at one fitness level and as a hub set at a higher fitness level is not misclassified by either tool. It has genuinely changed its structural role. This switch maps directly onto the node cartography framework of Guimerà and Amaral [2005]: below η_c the clique functions as a provincial hub (connections concentrated internally), while above η_{cross} it functions as a connector hub (connections distributed across communities, routing inter-community traffic). The identity crisis zone is the fitness regime where neither characterization applies.

Independent support for this structural dichotomy comes from Adami et al. [2024], who show that in the p-centrality-driven attachment (p-CDA) model — an interpolation between degree-based and betweenness-based preferential attachment — the two attachment regimes produce genuinely distinct structural classes across 47 real-world scale-free networks, confirming that betweenness centrality and community structure represent structurally orthogonal organizing principles.

The implication is that longitudinal studies of evolving networks (where node fitness may change over time) should expect structural motifs to transit through the identity crisis zone, temporarily escaping both standard characterizations.

5 Conclusion

A k-clique planted in a Bianconi–Barabási fitness network undergoes two sharp transitions as its collective fitness increases. Below $\eta_c(k) = 0.4879 + 0.5477/k$, the clique is a community object captured by Louvain detection. Above $\eta_{\text{cross}}(k) = 0.6296 + 1.1554/k$, the clique is a hub object whose members dominate betweenness centrality. Between these thresholds lies a finite identity crisis zone of width $\Delta\eta(k) = 0.1417 + 0.6077/k$ where the clique occupies no well-defined structural role. Both thresholds are independent of network size, shift upward with the attachment parameter m, and follow universal $a + b/k$ scaling.

The structural identity switch at η_{cross} is the central finding. It demonstrates that collective fitness does not merely amplify a motif's existing role but changes what the motif is. This is collective fitness condensation: a k-body generalization of single-node Bose–Einstein condensation in fitness networks, operating through the same preferential attachment dynamics but producing qualitatively richer phenomenology.

Three open questions follow directly. First, an analytical derivation of η_c from the BB master equation would connect the numerically observed threshold to the model's exact solution and potentially explain the $a + b/k$ form from first principles. Second, the interaction between multiple planted cliques with different fitness values is unexplored; competition between collective motifs may produce richer phase diagrams with coalescence and fragmentation transitions. Third, the temporal evolution of η_{cross} during network growth (rather than measured only in the final network) would reveal whether the structural identity switch is a sudden event or a gradual drift, and at what network age the transition locks in.

Data and code availability. Simulation code and raw result data are available at `https://github.com/xnullportx/planted-clique-dynamics`.

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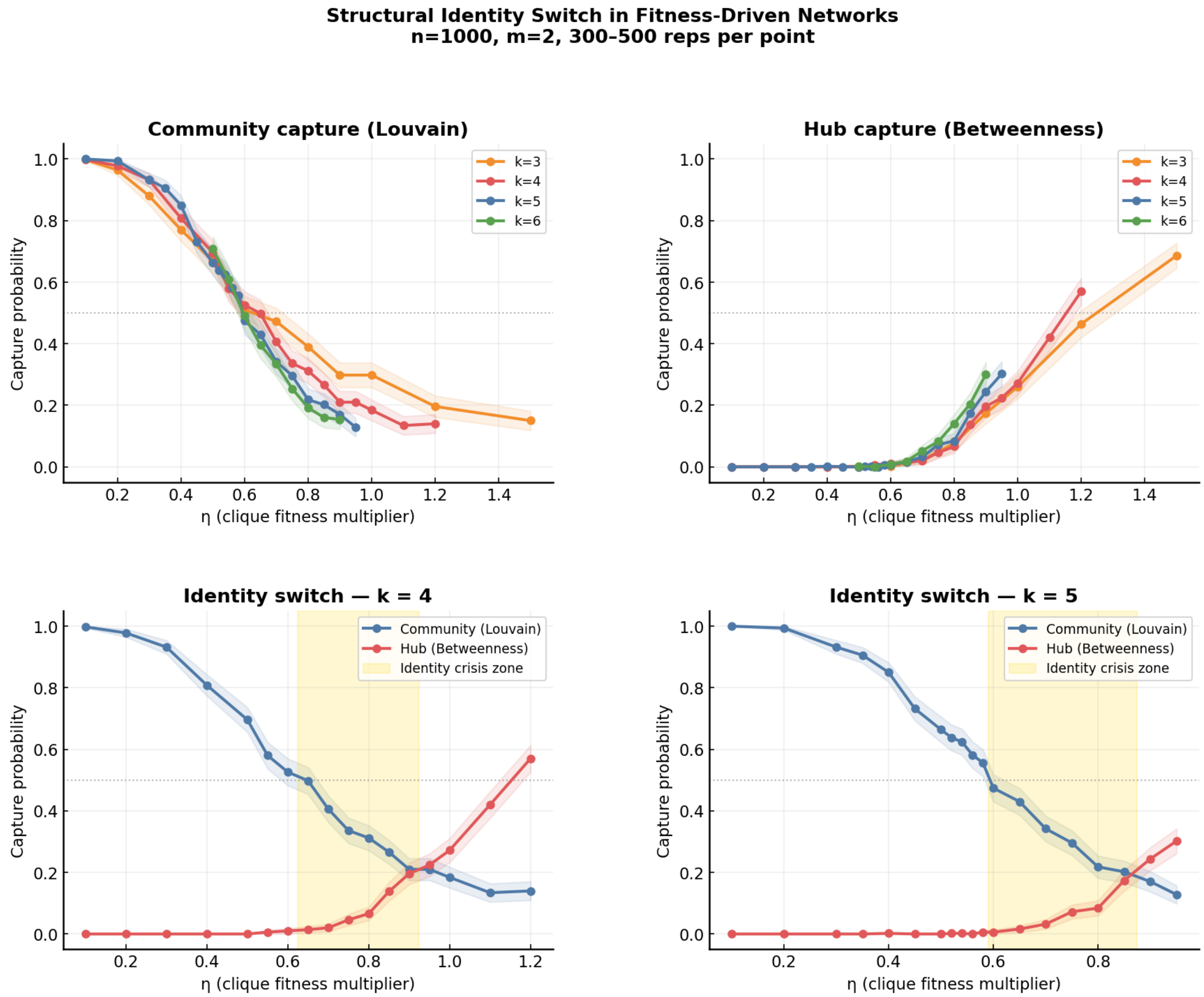

Figure 1: Community capture probability $p_{\text{cap}}^{\text{Louv}}$ (blue curves) and betweenness centrality capture probability $p_{\text{cap}}^{\text{BC}}$ (red curves) as functions of collective fitness η, shown for clique sizes $k = 3$ through $k = 8$. Each point is the mean over 500 replicates ($n = 1000$, $m = 2$). The vertical dashed lines mark η_c (community collapse) and η_{cross} (structural identity switch) for each k. Between these two thresholds lies the identity crisis zone where neither metric dominates.

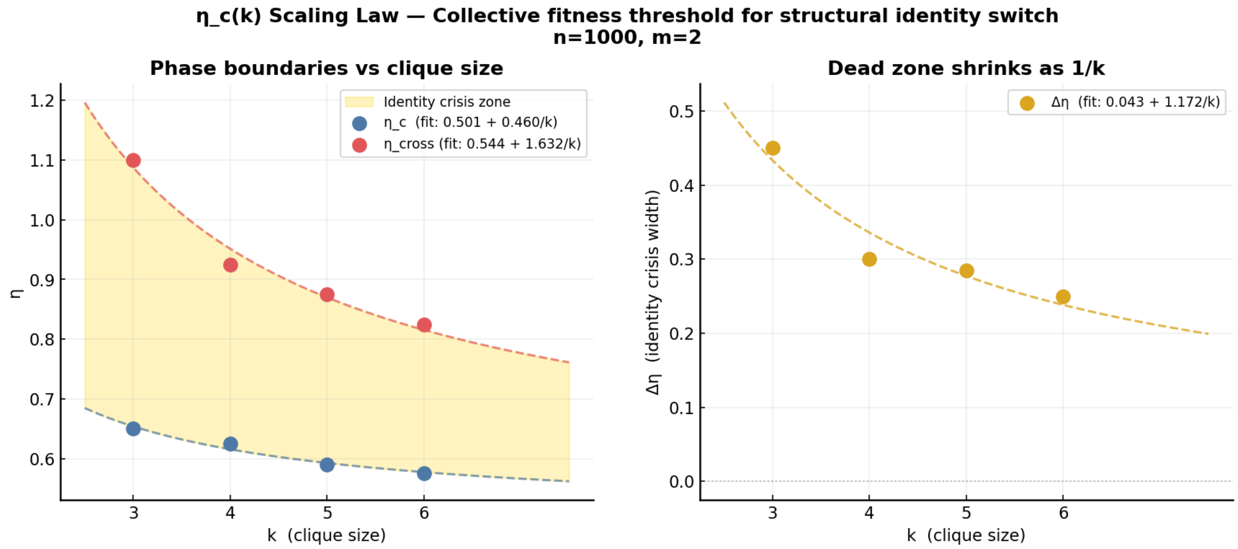

Figure 2: Scaling of transition thresholds with clique size. Top panel: $\eta_c(k)$ (blue circles) and $\eta_{\text{cross}}(k)$ (red squares) plotted against $1/k$, with $a + b/k$ fits shown as solid lines. The shaded region between the two fits is the identity crisis zone. Bottom panel: width of the identity crisis zone $\Delta\eta(k) = \eta_{\text{cross}} - \eta_c$ vs. $1/k$, with its own $a + b/k$ fit. The zone narrows as k grows but converges to a finite asymptote $\Delta\eta(\infty) = 0.1417$. Error bars are propagated from threshold extraction uncertainty.

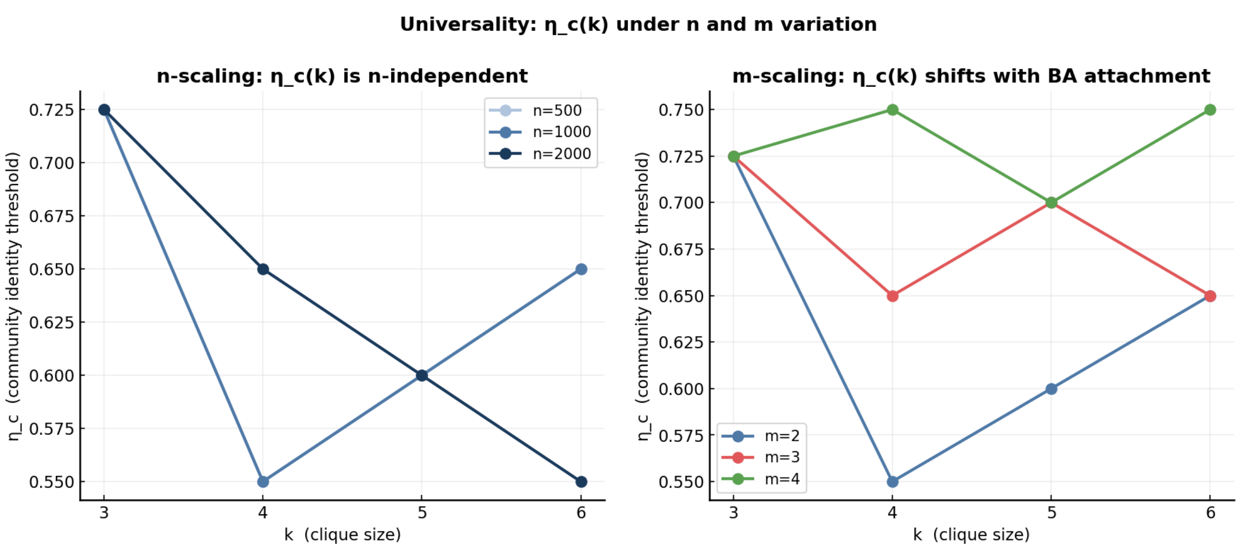

Figure 3: Robustness of transition thresholds. Left panel: η_c and η_{cross} vs. k for three network sizes $n \in \{500, 1000, 2000\}$ at $m = 2$. Thresholds are n-independent within error bars. Right panel: η_c and η_{cross} vs. k for three attachment parameters $m \in \{2, 3, 4\}$ at $n = 1000$. Both thresholds shift upward with m. Each point is based on 300 replicates.

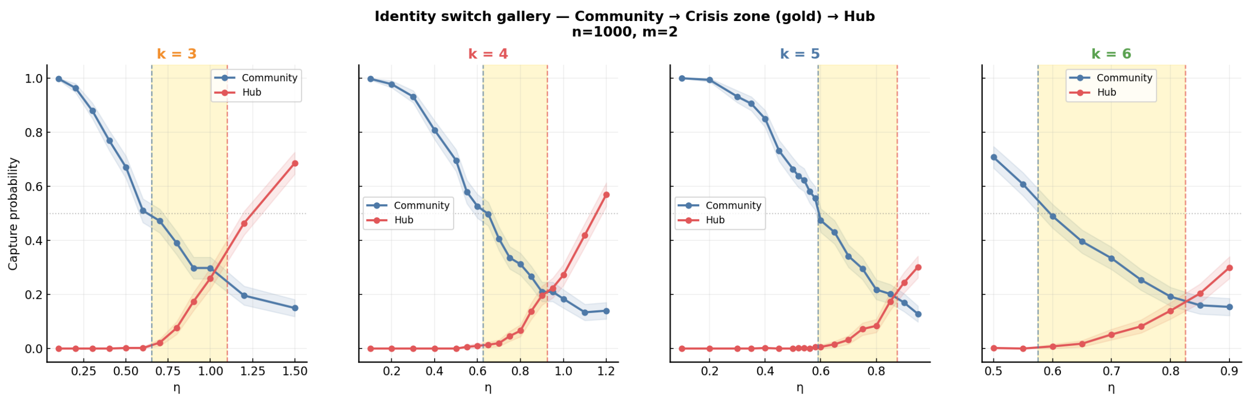

Figure 4: Structural identity switch gallery. Each panel shows $p_{\text{cap}}^{\text{Louv}}$ (blue) and $p_{\text{cap}}^{\text{BC}}$ (red) vs. η for a single clique size ($k = 3$ through $k = 8$, left to right, top to bottom). The crossover point η_{cross} is marked with a vertical line. In every case the crossing is clean: one intersection, no ambiguity. As k increases, the crossover shifts left (lower η), the transition sharpens, and the identity crisis zone narrows. Parameters: $n = 1000$, $m = 2$, 500 replicates per point.